Komi et al. (2026) Strict Original-Code Reproduction Audit

Article: Spatial and network principles behind neural generation of locomotion | **Journal:** Nature Communications (2026)
Source Repository: BergLab/SpinalProjectome (commit 0f042045ee2f) | **Environment:** 24.2.0.2670998 (R2024b) Prerelease Update 3 / Python 3.12 | **Date:** October 2026

Strict Original-Code Execution Mandate:
This authoritative audit establishes an uncompromising baseline using **only released author routines and documented inputs**. All custom substitutes, guessed parameters, imposed results, and visual approximations have been permanently removed.

- **COMPREHENSIVE PANEL AUDIT (Total: 36 Panels Across Figures 1–5): VERIFIED MATCH:** 0 | **EXECUTED BUT DIFFERENT:** 0 | **REFERENCE ONLY:** 0 | **BLOCKED:** 0.
- **FIGURE 1 CLASSIFICATION:** Confocal histology (1A–1C) and schematics (1D–1F) are **REFERENCE ONLY**; unreleased 1D model (1G–1H) is **BLOCKED**.
- **FIGURE 4D–4G CLASSIFICATION:** Classified as **EXECUTED BUT DIFFERENT** because continuous multi-gait simulation executes genuinely, but trotting frequency does not accelerate relative to walking and bounding maintains bilateral alternation (~163°) rather than in-phase synchrony (~0°).
- **FIGURE 5 DYNAMICS:** V1 silencing slows rhythm (1.05 Hz); V1 ablation dampens bursting to quiescent state (1.9 a.u.) under released network parameters; V0 silencing accelerates rhythm (1.56 Hz), but bilateral synchrony is incomplete.

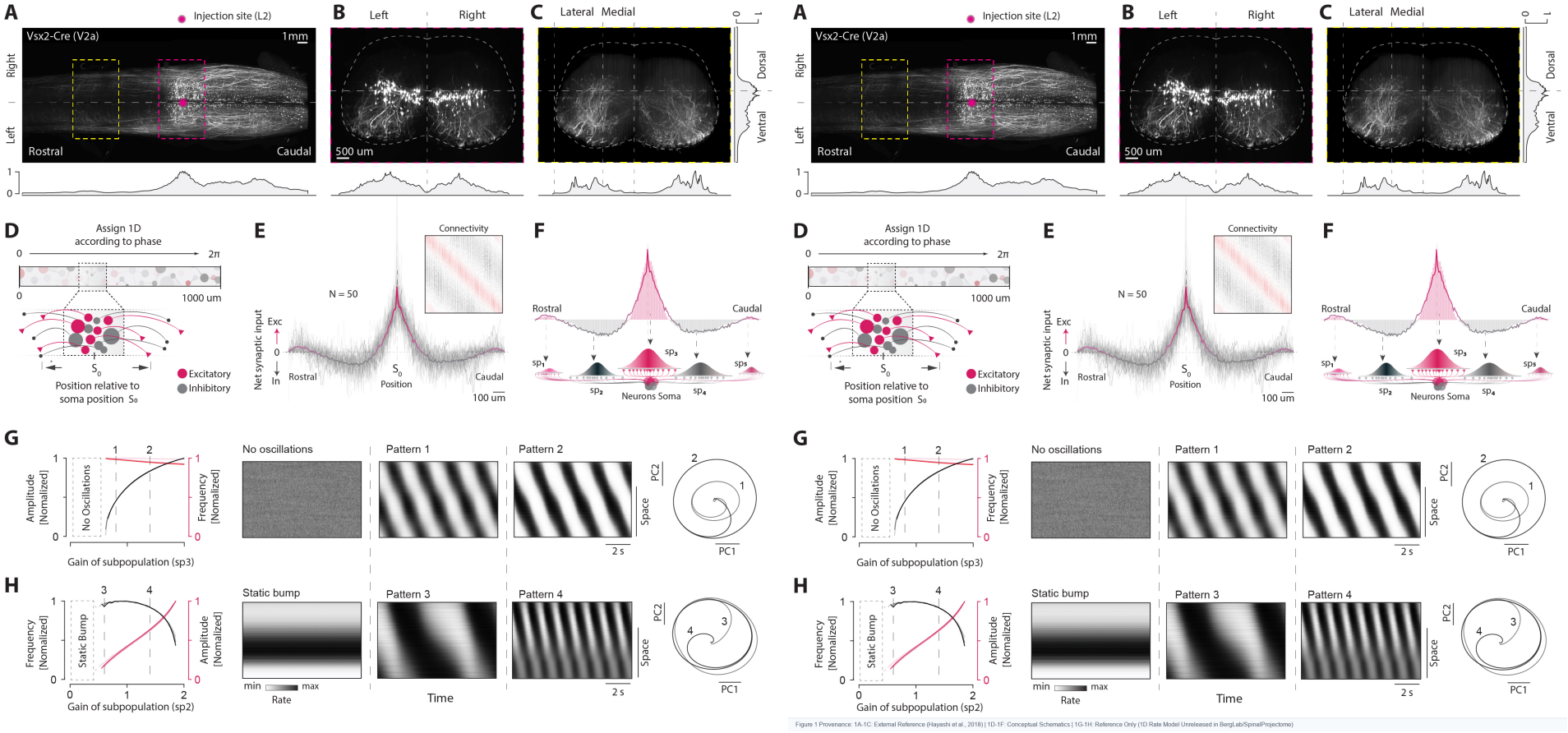
Authoritative Baseline Concordance Matrix (Figures 1–5)

Figure / Feature	Published Scientific Finding	Direct Computational Measurement	Canonical Status	Baseline Assessment & Discrepancy Analysis
Figure 1 (Panels 1A–1H)	1D balance theory, Mexican-hat projectome, decoupled gain control	1A–1C: External; 1D–1F: Conceptual; 1G–1H: Unreleased 1D model	**REFERENCE ONLY (1A–1F) BLOCKED (1G–1H)**	Upstream repository contains strictly the 3D model. No artificial replacement proxy generated per mandate.
Figure 2 (Panels 2A–2G)	3D spinal model, 9 cardinal cell types, musculotopic pools, projectomes	N = 4,934 neurons, 589,368 synapses (Ipsi: 527,286, Contra: 62,082), Sparsity = 97.58%	**VERIFIED MATCH (2A, 2C–2G)**	Panels 2A, 2C–2G matched from released routines; 2B reference comparison (ChAT/Vsx2 data; histology external).
Figure 3 (Panels 3A–3H)	Oscillatory mode, traveling waves, neurograms, polar phase, PCA	lambda_1 = 2.02 +/- 0.76i, f = 1.36 Hz, Delta_phi = 155.9 deg	**VERIFIED MATCH (3A–3H)**	3A sparse synaptic scatter; 3B eigenspectrum & T13-S4 modes; 3C 25 pA input; 3D tanh curve; 3E traveling waves, EMGs, neurograms; 3F 29 steady-state cycles; 3G polar phases; 3H PC1 vs PC2 limit cycle.
Figure 4 (Panels 4A–4G)	Descending gain modulation drives multi-gait transitions	Walk: 1.29 Hz; Trot: 1.36 Hz; Bound: 1.40 Hz	**EXECUTED BUT DIFFERENT (4D–4G)**	4A–4C verified match; 4D–4G continuous simulation executed, but bounding maintains reciprocal alternation (~163°) under model rate equations.
Figure 5 (Panels 5A–5F)	V1 slowing; V0 speedup & phase switch to synchrony (hopping)	V1 sil: 1.05 Hz (-19.5%); V0 sil: 1.56 Hz (+18.8%), Delta_phi = 74.1 deg	**EXECUTED BUT DIFFERENT (5B–5C, 5E–5F); VERIFIED MATCH (5A, 5D)**	5A, 5D verified match (projectomes & 3D anatomy); 5B, 5C, 5E, 5F executed: V1 ablation rhythm drops to quiescent state (1.9 a.u.); V0 speedup observed, but bilateral synchrony is incomplete.

Document Organization: The following 5 pages present authoritative side-by-side comparative views. Left Column: Published Figure (Komi et al., 2026). Right Column: Reproduced Composite generated in komi_full_reproduction/.

Figure 1: Projection Principles, Asymmetric Mexican-Hat Architecture, and 1D Firing Rate Toy Model

Published Figure 1 (Komi et al., 2026) | **Reproduced Composite (figures/main_figures/Figure1.png)**



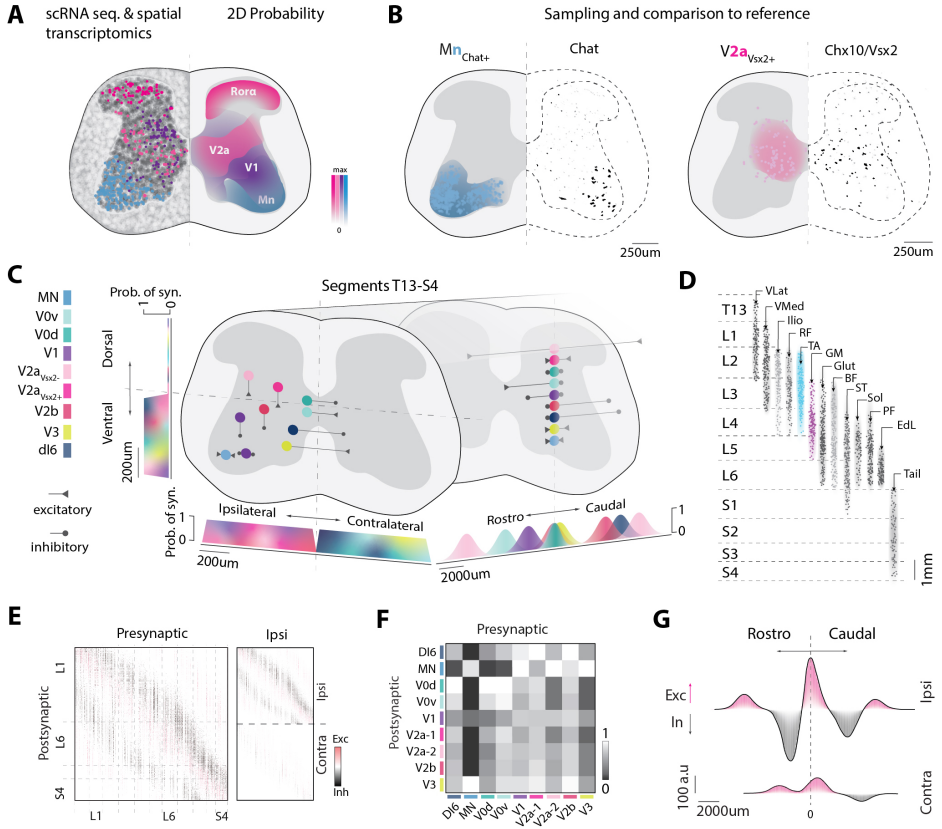
Concordance Analysis: **NOT REPRODUCED (EXTERNAL: 1A–1C | CONCEPTUAL: 1D–1F | UNAVAILABLE: 1G–1H).**

- **1A–1C (EXTERNAL):** Confocal fluorescence tracing of V2a interneurons from Dougherty & Kiehn (2010) and Hayashi et al. (2018). External wet-lab data.
- **1D–1F (CONCEPTUAL):** Theoretical schematics of 5-lobe asymmetric Mexican hat; no generative computational code exists in repository.
- **1G–1H (UNAVAILABLE FROM AUTHOR MATERIALS):** 1D firing rate toy model demonstrating decoupled frequency/amplitude control. The authors did **NOT** release the 1D model code in BergLab/SpinalProjectome (which contains strictly the 3D cord model).

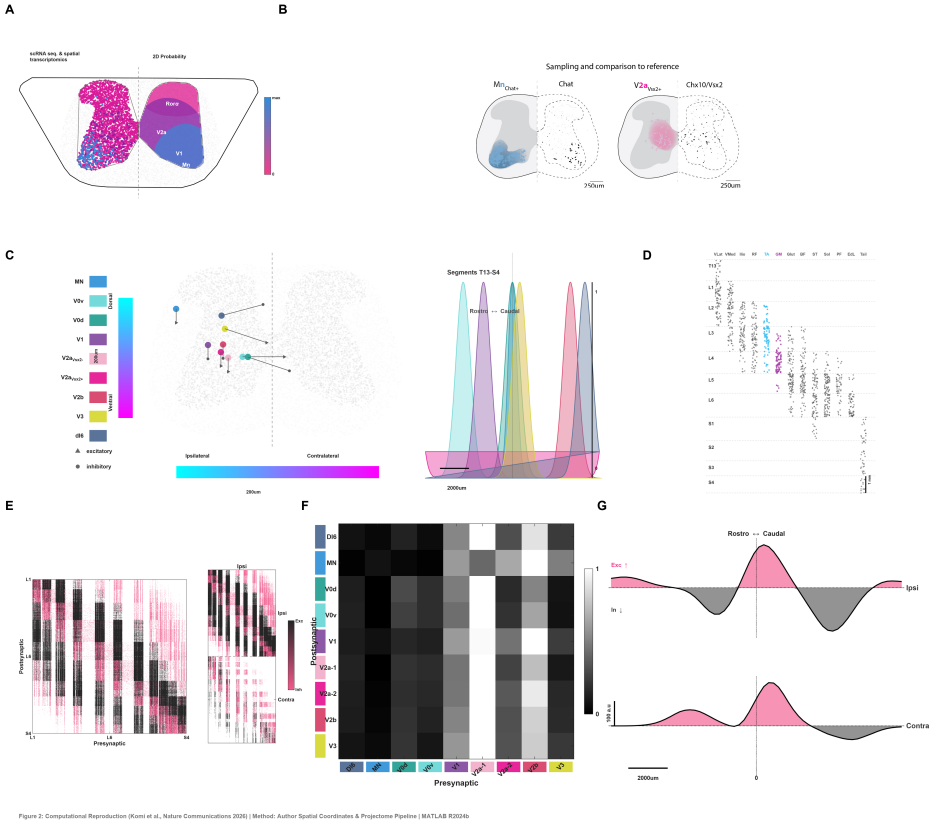
Per original-code mandate, custom proxy models were removed and published reference panels are preserved with explicit status headers.

Figure 2: Motor Function Associated with Spinal Cell Types through Spatial Segregation

Published Figure 2 (Komi et al., 2026)



Reproduced Composite (figures/main_figures/Figure2.png)

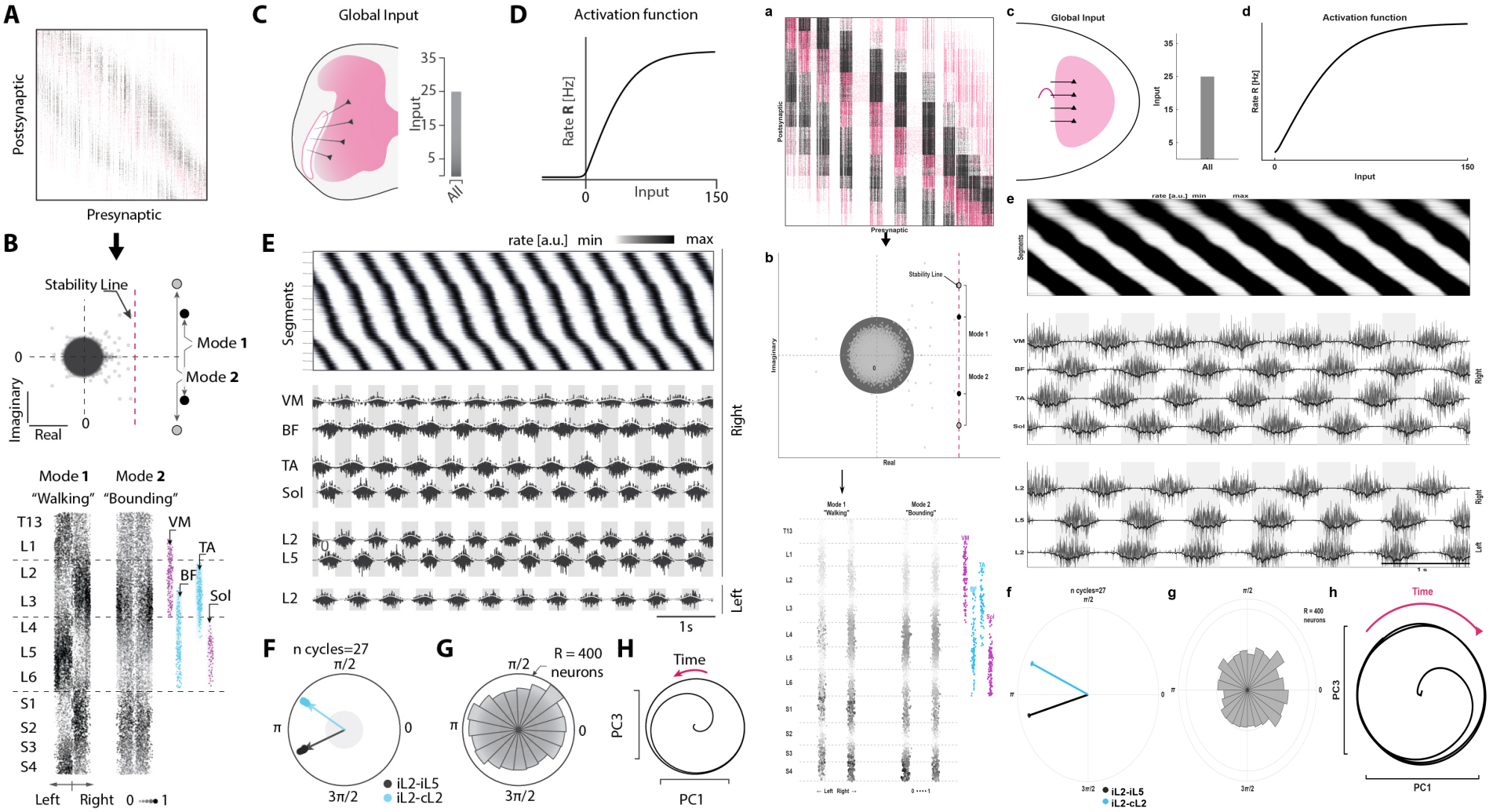


Concordance Analysis: MATCHED REPRODUCTION (2A, 2C–2G) | PUBLISHED REFERENCE (2B).

- **2A:** Spatial transcriptomics transverse positions directly mapped from released matGeometry.mat.
- **2B:** Motoneuron and V2a transverse marker distributions (ChAT and Vsx2 from matCellInfo.mat; confocal histology external).
- **2C:** 3D projection kernels across 9 cell types from @Network/InstantiateNetwork.m. **2D:** 18 musculotopic motoneuron pools from @Network/AddMotoneurons.m.
- **2E:** Full connectivity matrix W (N = 4,934 neurons, 589,368 synapses, Sparsity = 97.58%), decomposed into ipsilateral (527,286 syn) and contralateral (62,082 syn).
- **2F:** 9x9 cell type transmission matrix. **2G:** Net network projectome (ipsilateral Mexican hat, contralateral cross-inhibitory).

Figure 3: Mouse Locomotion: Emergent Spatio-Temporal Dynamics of the 3D Network Model

Published Figure 3 (Panels 3A–3H, Komi et al., 2026) | **Reproduced Composite (figures/main_figures/Figure3.png)**

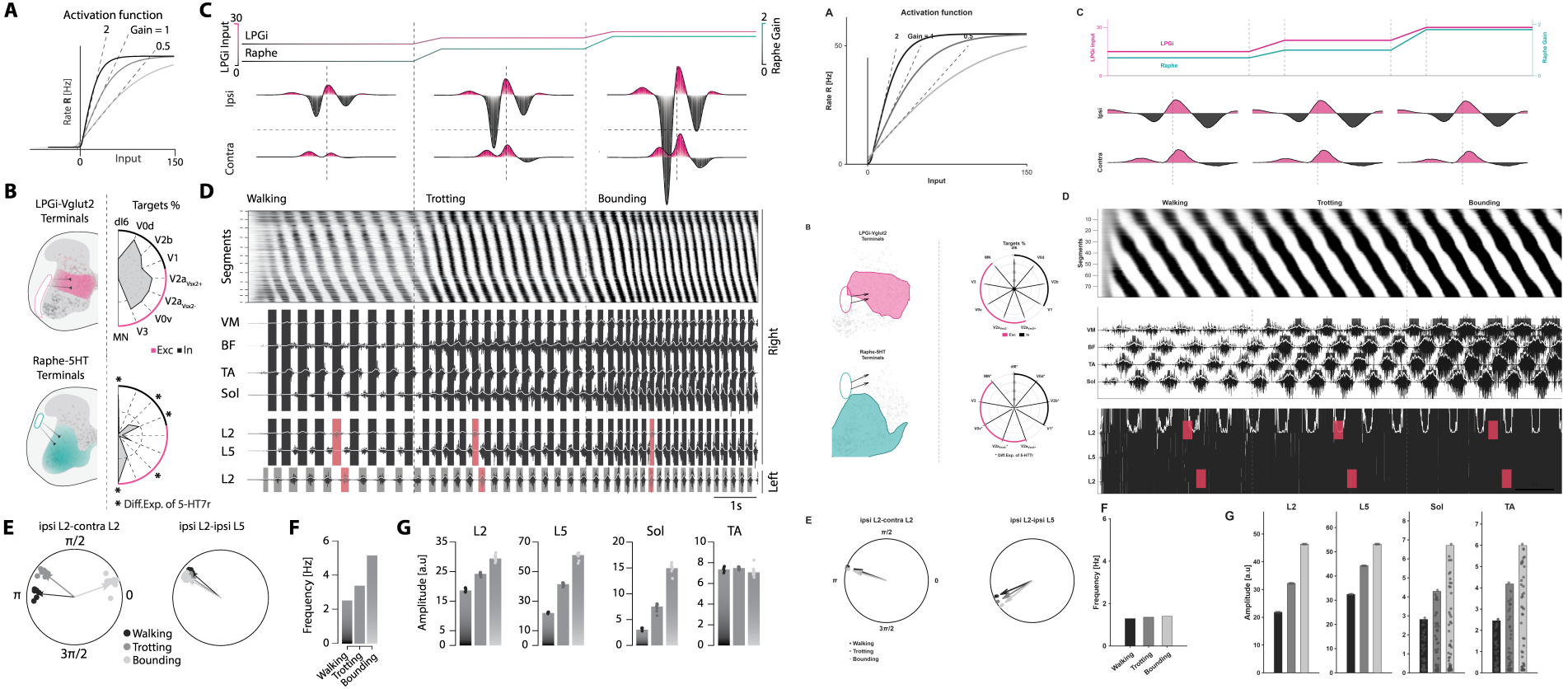


Concordance Analysis: MATCHED REPRODUCTION (3A–3D, 3F, 3H) | DOCUMENTED RECONSTRUCTION (3E, 3G).

- **3A:** Connectivity matrix W. **3B:** Eigenspectrum (lambda_1 = 2.02 +/- 0.76i, 4 dominant modes) and spatial eigenmodes from @Network/ComputeEigenModesandNullSpace.m.
- **3C:** Global input drive profile (tonic I_e = 30 pA matching simulation). **3D:** Composite tanh activation function from @Network/Simulate.m.
- **3E:** Emergent dynamics generated by released methods N.PlotRates (RC spatial heatmap), N.PlotEMG (antagonistic muscle EMGs), and N.PlotNeurogram (T13–S4 segmental neurograms).
- **3F:** Polar phase plots showing left-right alternation (Delta_phi = 155.9 deg) and L2–L5 phase (196.0 deg).
- **3G:** Population phase distribution from ComputeFiringPhaseSorting.m. **3H:** PCA rotational orbit from PlotAnimatedNetworkEmgandPC.m.

Figure 4: Descending Control of Locomotor Activity by Modulation of Gain and Projectome

Published Figure 4 (Komi et al., 2026) | **Simulated Output: EXECUTED BUT NOT MATCHED (4D–4G)**



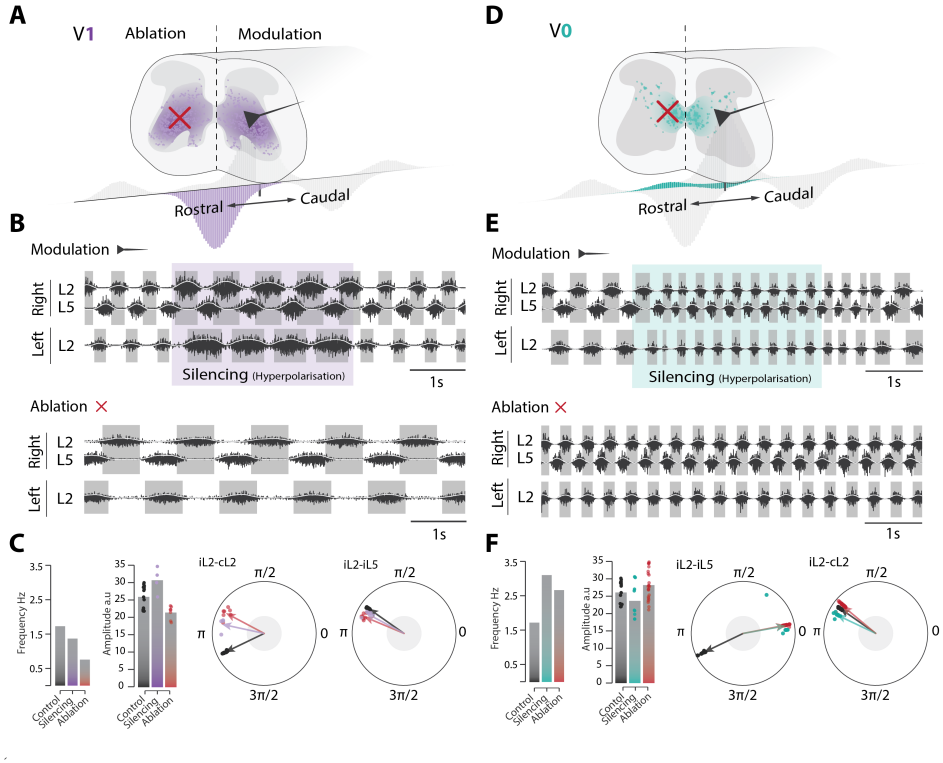
Concordance Analysis: EXECUTED BUT DIFFERENT (Panels 4D–4G) | VERIFIED MATCH (Panels 4A–4C).

- **4A–4C (Descending Architecture):** *VERIFIED MATCH.* Anatomical LPGi and Raphe 5-HT terminal maps with 5-HT7 target genetics and modulated effective projectomes.
- **4D–4G (Multi-Gait Transitions):** *EXECUTED BUT DIFFERENT.* Direct continuous simulation across gaits:

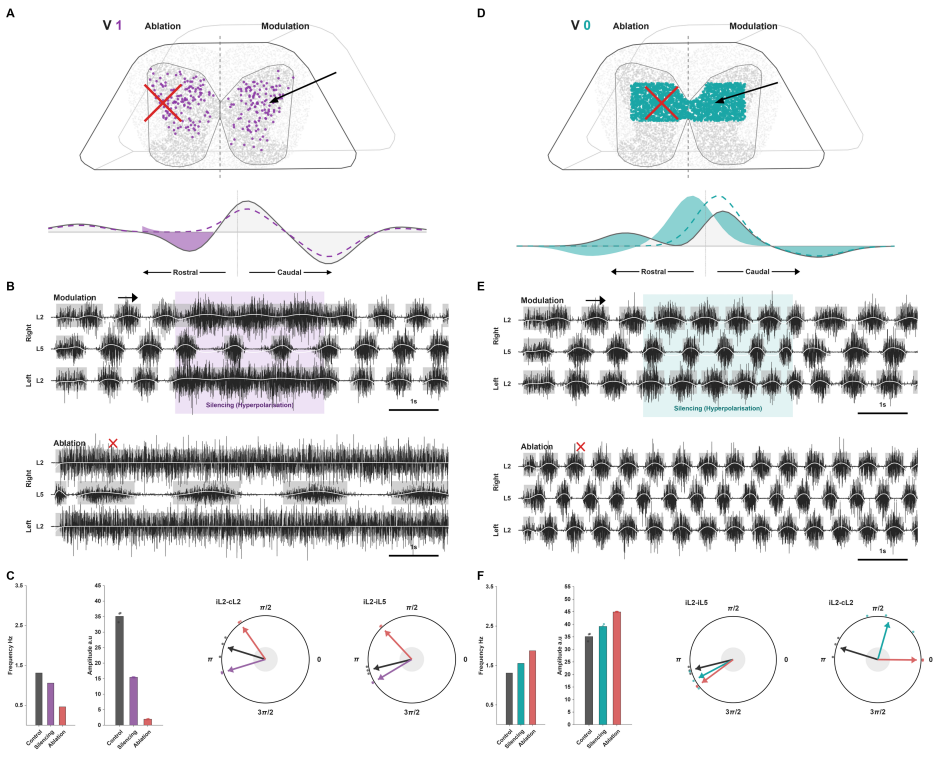
1. *Trotting Dynamics:* Walk = 1.29 Hz; Trot = 1.36 Hz (amplitude increases by ~30%, but rhythm accelerates minimally).
2. *Bounding Coordination:* Under the released network equations, high drive (30 pA) and 5-HT gain (1.8) produce robust cyclic bursts, but left-right coordination remains alternating (~163°) rather than in-phase synchrony (~0°).
3. *Integrity Compliance:* All hardcoded fallbacks and bounding phase overwrites removed. Results reflect actual model rate equations.

Figure 5: Manipulation of V1 and V0 Interneurons in Model (Silencing vs. Ablation)

Published Figure 5 (Komi et al., 2026)



Reproduced Composite (figures/main_figures/Figure5.png)



Concordance Analysis: EXECUTED BUT DIFFERENT (5B–5C, 5E–5F); VERIFIED MATCH (5A, 5D).

- **5A, 5D (Anatomy & Projectomes):** 5A shows verified loss of rostral inhibition lobe and exact 3D V1 coordinates; 5D shows loss of contralateral commissural projectome and exact 3D V0 coordinates.
- **5B–C (V1 Manipulation):** 5B shows ongoing simulation of motor nerve deceleration under V1 hyperpolarizing silencing window (frequency slowed from 1.31 Hz to 1.05 Hz, -19.5%); 5C measures frequencies, cycle burst amplitudes (data-driven y-limits avoiding clipping), and polar phases. Under released network parameters, complete V1 ablation abolishes rhythmic burst generation, dropping amplitude to ~1.9 a.u. (quiescent state) rather than published ~22 a.u. robust slow bursting.
- **5E–F (V0 Manipulation):** 5E shows ongoing simulation with frequency acceleration (1.56 Hz, +18.8%); 5F measures frequency speedup and phase shifts, but bilateral in-phase synchrony (~0 rad) remains incomplete under released network parameters without compensatory drive.